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Bias-Free Langua

Contents

Introduction

This document describes how to configure and verify e- Border Gateway Protocol (eBGP) peering between vPC Nexus pair and another device. The configuration on the external device is shown as Cisco CLI NX-OS for clarity.

Prerequisites

Requirements

Cisco recommends that you have knowledge of these topics:

- vPC operation and configuration basic concepts. For reference: https://www.cisco.com/c/dam/en/us/td/docs/switches/datacenter/sw/design/vpc_design/vpc_best_practices_design_guide.pdf
- BGP operation and configuration

Components Used

The information in this document is based on these software and hardware versions:

leaf1#	C93108TC-FX	NXOS 9.3(3)
leaf2#	C93108TC-FX	NXOS 9.3(3)
ExternalDevice	N9K-C9396PX	NXOS: version 9.2(3)

The information in this document was created from the devices in a specific lab environment. All of the devices used in this document started with a cleared (default) configuration. If your network is live, ensure that you understand the potential impact of any command.

Background Information

Routing protocols (OSPF, ISIS, RIP, EIGRP, BGP) peering between vPC pair and an external device. This is supported as per: https://www.cisco.com/c/en/us/support/docs/ip/ip-routing/118997-technote-nexus-00.html. This article describes additional notes and configuration examples for eBGP as a routing protocol.

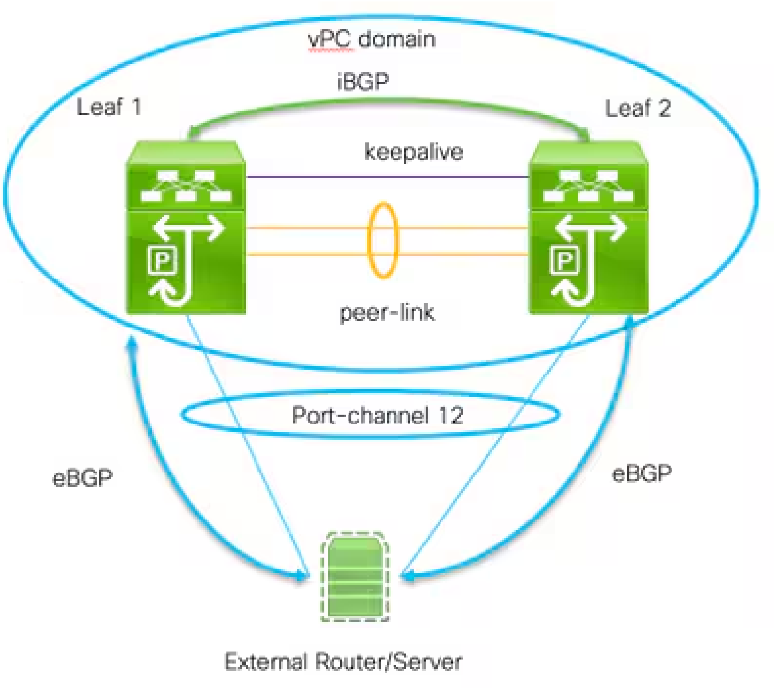
Several problems are posed when trying to enable routing protocol peering over a vPC, which are not existent when using

standard port-channel:

1. It is not deterministic which member of the port-channel the External Device will use to forward traffic for each mac-address. It is possible that the External Device will send BGP packets for Leaf-1 over the link to Leaf-1. When Leaf-1 receives it, it will punt it to the CPU and discard it (not own ip address), thus BGP (unicast) or other protocol (multicast) will be flapping constantly. Here helps the command peer-gateway.
2. Even with peer-gateway command, TTL will be decreased on such a packet. THe new command in NX-OS - **layer3 peer-router** disables that.
3. iBGP between the two vPC members is needed to obey the BGP rule for iBGP between all neighbors. We are running inside vrf on vPC side, thus only those two members need to run iBGP. THis is also needed in case of link failiure to rest of the NX-OS network (VXlan or other) and provide redundancy.

Configure

Network Diagram



Two important commands are needed to enable this peering:

- **Peer-gateway** - The vPC peer-gateway functionality allows a vPC switch to act as the active gateway for packets that are addressed to the router MAC address of the vPC peer
- **layer3 peer-router -** no change for TTL of packets destined to the peer, external device sees the vPC domain as single physical entity from layer route protocol peering perspective as well.

Configurations

```
Leaf 1:
! Form the vPC domain:
vpc domain 1
  peer-switch
  role priority 10 peer-keepalive destination 192.0.2.2 source 192.0.2.1 peer-
gateway layer3 peer-router ipv6 nd synchronize ip arp synchronize
!
!vPC peer-link interface members
interface Ethernet1/53 - 54
  description vPC-Peerlink member
  switchport
  switchport mode trunk
  channel-group 11 mode active
  no shutdown
!
! vPC peer-link port-channel
interface port-channel11
  description vPC-peerlink
  switchport
  switchport mode trunk
  spanning-tree port type network
  no shutdown
  vpc peer-link
!
! vPC port-channel member to External Device
interface Ethernet1/52
  description ExternalDevice Eth2/13
  switchport
  switchport mode trunk
  switchport trunk allowed vlan 203,205
  mtu 9216
  channel-group 12 mode active
  no shutdown
!
! vPC port-channel to External Device
interface port-channel12
  description vPC port-channel to External Device
  switchport
  switchport mode trunk
  switchport trunk allowed vlan 203,205
  mtu 9216
  vpc 12
!
! Layer 3 interface to the Eternal device:
interface Vlan205
```

```
  no shutdown
  vrf member Customer
! BFD for eBGP
  bfd interval 500 min_rx 500 multiplier 3
! Disable bfd echo, as it is not supported over vPC
  no bfd echo
  no ip redirects
! We use/29 as we need 3 ip address, one per each member of the ! vPC domain and
  ip address 198.51.100.1/29 tag 800204
! Disable redirects - this is needed to enable BFD
  no ipv6 redirects
!
router bgp 65535
router bgp 65535
  router-id 203.0.113.1
  log-neighbor-changes
  address-family ipv4 unicast
   Customer router-id 198.51.100.1 address-family ipv4 unicast neighbor 198.51.1
! Form the vPC domain:
vpc domain 1
  peer-switch
  role priority 10
  peer-keepalive destination 192.0.2.1 source 192.0.2.2
  peer-gateway
  layer3 peer-router
  ipv6 nd synchronize
  ip arp synchronize
!
!vPC peer-link interface members
interface Ethernet1/53 - 54
  description vPC-Peerlink member
  switchport
  switchport mode trunk
  channel-group 11 mode active
  no shutdown
!
! vPC peer-link port-channel
interface port-channel11
  description vPC-peerlink
  switchport
  switchport mode trunk
  spanning-tree port type network
  no shutdown
  vpc peer-link
!
! vPC port-channel member to External Device
interface Ethernet1/52
  description ExternalDevice Eth2/13
```

```
  switchport
  switchport mode trunk
  switchport trunk allowed vlan 203,205
  mtu 9216
  channel-group 12 mode active
  no shutdown
!
! vPC port-channel to External Device
interface port-channel12
  description vPC port-channel to External Device
  switchport
  switchport mode trunk
  switchport trunk allowed vlan 203,205
  mtu 9216
  vpc 12
!
! Layer 3 interface to the Eternal device:
interface Vlan205
  no shutdown
  vrf member Customer
! BFD for eBGP
  bfd interval 500 min_rx 500 multiplier 3
! Disable bfd echo, as it is not supported over vPC
  no bfd echo
  no ip redirects
! We use/29 as we need 3 ip address, one per each member of the ! vPC domain and
  ip address 198.51.100.2/29 tag 800204
! Disable redirects - this is needed to enable BFD
  no ipv6 redirects
!
router bgp 65535
router bgp 65535
  router-id 203.0.113.2
  log-neighbor-changes
  address-family ipv4 unicast
 vrf Customer
    router-id 198.51.100.2
    address-family ipv4 unicast
    neighbor 198.51.100.1
      description Leaf-2
      remote-as 65535
      address-family ipv4 unicast
        soft-reconfiguration inbound always
    neighbor 198.51.100.3
      description to External Device
      bfd
      remote-as 65000
      update-source Vlan205
```

```
      address-family ipv4 unicast
        soft-reconfiguration inbound always
!
External Device (NX-OS style CLI):
interface Ethernet2/13 - 14
  switchport
  switchport mode trunk
  switchport trunk allowed vlan 203,205
  mtu 9216
  channel-group 12 mode active
  no shutdown
!
interface port-channel12
  switchport
  switchport mode trunk
  switchport trunk allowed vlan 203,205
  mtu 9216
  no shutdown
!
interface Vlan205
  no shutdown
  mtu 9216
! See notes in Leaf-1 and Leaf 2 for BFD
  bfd interval 500 min_rx 500 multiplier 3
  no bfd echo
  no ip redirects
  ip address 198.51.100.3/29
  no ipv6 redirects
!
router bgp 65000
  log-neighbor-changes
  address-family ipv4 unicast
   neighbor 198.51.100.1 remote-as 65535
      description to Leaf-1
      update-source Vlan205
      bfd
   neighbor 198.51.100.2 remote-as 65535
      description to Leaf-2
      update-source Vlan205
      bfd
end
!
```

Verify

Below is output of show bgp ipv4 unicast neighbors. It verifies that:

1. BGP neighborship is established and stable

2. BFD is enabled between external neighbors

```
Leaf 1/2:
show bgp ipv4 unicast neighbors vrf Customer
BGP neighbor is 203.0.113.2, remote AS 65535, ibgp link, Peer index 4
  BGP version 4, remote router ID 203.0.113.2
  Neighbor previous state = OpenConfirm
  BGP state = Established, up for 6d22h
  Neighbor vrf: Customer
  Peer is directly attached, interface Vlan205
  Last read 00:00:14, hold time = 180, keepalive interval is 60 seconds
  Last written 00:00:03, keepalive timer expiry due 00:00:56
  Received 10012 messages, 0 notifications, 0 bytes in queue
...
BGP neighbor is 203.0.113.2.3, remote AS 65000, ebgp link, Peer index 3
  BGP version 4, remote router ID 203.0.113.2
  Neighbor previous state = OpenConfirm
  BGP state = Established, up for 1d00h
  Neighbor vrf: Customer
  Using Vlan205 as update source for this peer
  Peer is directly attached, interface Vlan205
  BFD live-detection is configured and enabled, state is Up
  Last read 00:00:22, hold time = 180, keepalive interval is 60 seconds
  Last written 00:00:56, keepalive timer expiry due 00:00:03

!
External Device:
show bgp ipv4 unicast neighbors
BGP neighbor is 203.0.113.1, remote AS 65535, ebgp link, Peer index 3
  Inherits peer configuration from peer-template Cust_BGP_Peer
  BGP version 4, remote router ID 203.0.113.1
  BGP state = Established, up for 1d00h
  Peer is directly attached, interface Vlan205
  Enable logging neighbor events
  BFD live-detection is configured and enabled, state is Up
  Last read 0.660288, hold time = 180, keepalive interval is 60 seconds
  Last written 00:00:26, keepalive timer expiry due 00:00:33
  Received 10122 messages, 1 notifications, 0 bytes in queue
  Sent 10086 messages, 1 notifications, 0(0) bytes in queue
  Connections established 14, dropped 13
  Last reset by us 1d00h, due to bfd session down
  Last reset by peer 6d22h, due to other configuration change
....
```

Troubleshoot

Following commands wil lhelp verify operation:

```
show vpc
show vpc consistency-parameters global
show vpc consistency-parameters interface <interface>
show bgp ipv4 unicast neighbors
show bgp ipv4 unicast summary
```

Revision History

Revision	Publish Date	Comments
1.0	19-May-2021	Initial Release